



DSBA 6165 — DEEP LEARNING PROJECT PROPOSAL

AI Regulatory Risk Detection & Evidence Retrieval Using Deep Learning

Classifying AI systems into EU AI Act risk tiers with rule-aware transformers

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Team of 2

Classifying AI systems under the EU AI Act is manual, slow, and inconsistent

The EU AI Act sorts every AI system into one of four risk tiers. Today, legal experts read dense system descriptions and interpret them against equally dense legal text — by hand, one system at a time.

- **Slow** Days per system to classify
- **Expensive** Requires specialized legal expertise
- **Inconsistent** Interpretations vary between reviewers
- **Not scalable** Thousands of systems need review

The opportunity

An automated classifier won't replace legal judgment — but it can triage systems fast, flag likely Prohibited/High-risk cases for expert review, and cite the article behind each call.

WHY IT MATTERS

€35M

or 7% of global annual turnover — maximum fine for non-compliance

First of its kind

World's first comprehensive AI regulation — sets a global precedent

Open research problem

The Act entered into force in 2024, but high-risk obligations phase in through 2026-27 — automated compliance tooling is still emerging.

Three papers frame the gap this project fills

Benchmark

Davvetas et al., 2026

AI Act Evaluation Benchmark

First open benchmark for evaluating NLP/RAG systems against the EU AI Act.

Gap: evaluates systems, but proposes no classification model of its own.

Method

Meghdadi & Inkpen, 2024

Transformer-Based Legal Classification

RoBERTa + CNN reaches 86-88% on legal violation detection and inference tasks (LegalLens-2024).

Gap: general legal violations — not applied to the AI Act specifically.

Method

Ali et al., 2025

Hybrid Rule-Based + XAI Compliance

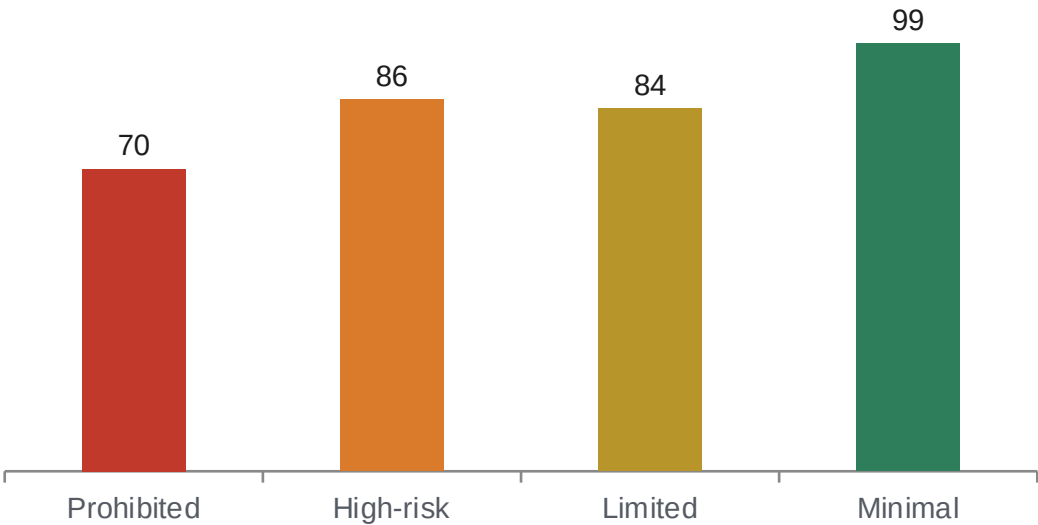
Fusing symbolic rules with contextual AI models reaches 97.9% accuracy on GDPR/ePrivacy compliance, with article-level explainability.

Gap: built for GDPR/ePrivacy — not tested on the AI Act's risk tiers.

Takeaway: BERT works well on legal language, and rule + ML hybrids help — but no one has combined both for the AI Act. That's the gap this project targets.

AI Act Evaluation Benchmark, complemented with the full Act text

Scenarios by Risk Tier



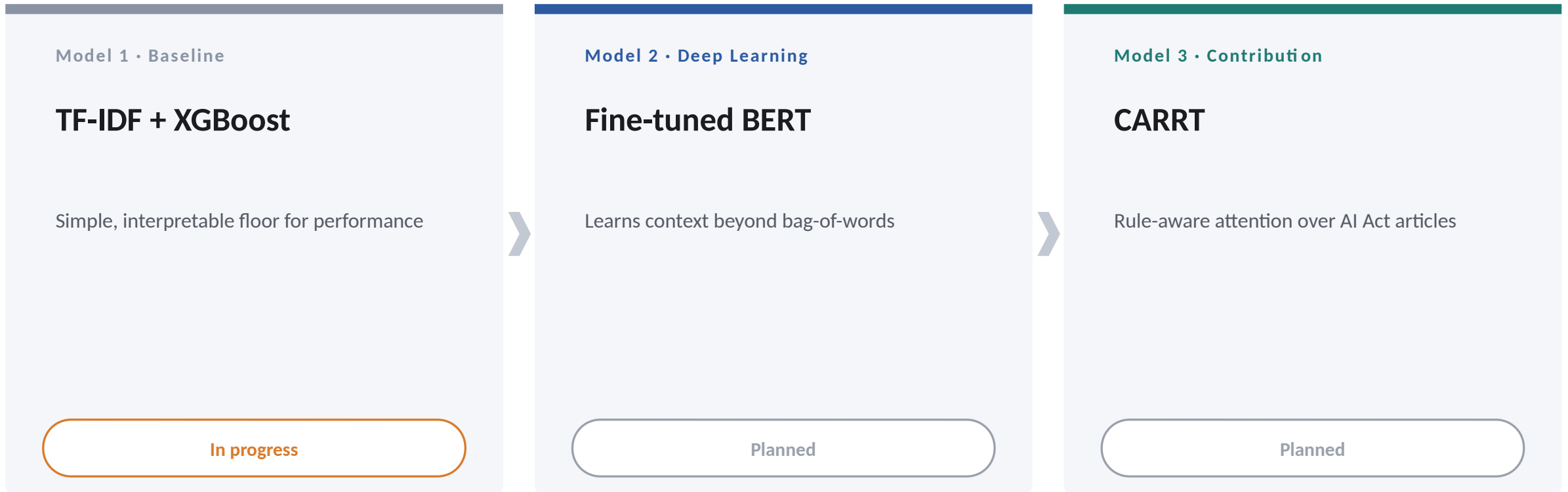
Total: 339 labeled scenarios · train/test split 271 / 68 (80/20)

Source	Contents
AI Act Evaluation Benchmark (2026)	339 labeled scenarios · role, intended use, system type, domain, related articles, obligations · CC-BY 4.0
EU AI Act full text (EUR-Lex)	933 pre-processed, embedded law-text chunks · used for evidence retrieval & rule injection

Note: a known label-ambiguity gap, not a flaw

Scenarios are generated per risk tier by an LLM built on a legal-domain framework, not annotated case-by-case by lawyers — cite it that way. The paper's own qualitative checks also found scenario ambiguity is more common in Prohibited/High-risk cases than Limited/Minimal — exactly the tiers CARRT targets, so this is a real challenge to design for, not just noise.

Three models, increasing sophistication



Every model outputs a risk-tier classification plus a retrieved supporting article — classification and explanation together.

CARRT: Constraint-Aware Regulatory Risk Transformer

Standard BERT treats every word the same. CARRT adds a second signal — which AI Act articles are relevant to this system description — and lets that signal bias the model's attention toward legally decisive language. The goal is to catch high-stakes cases (Prohibited, High-risk) that plain BERT tends to miss.

- **Why it's novel**

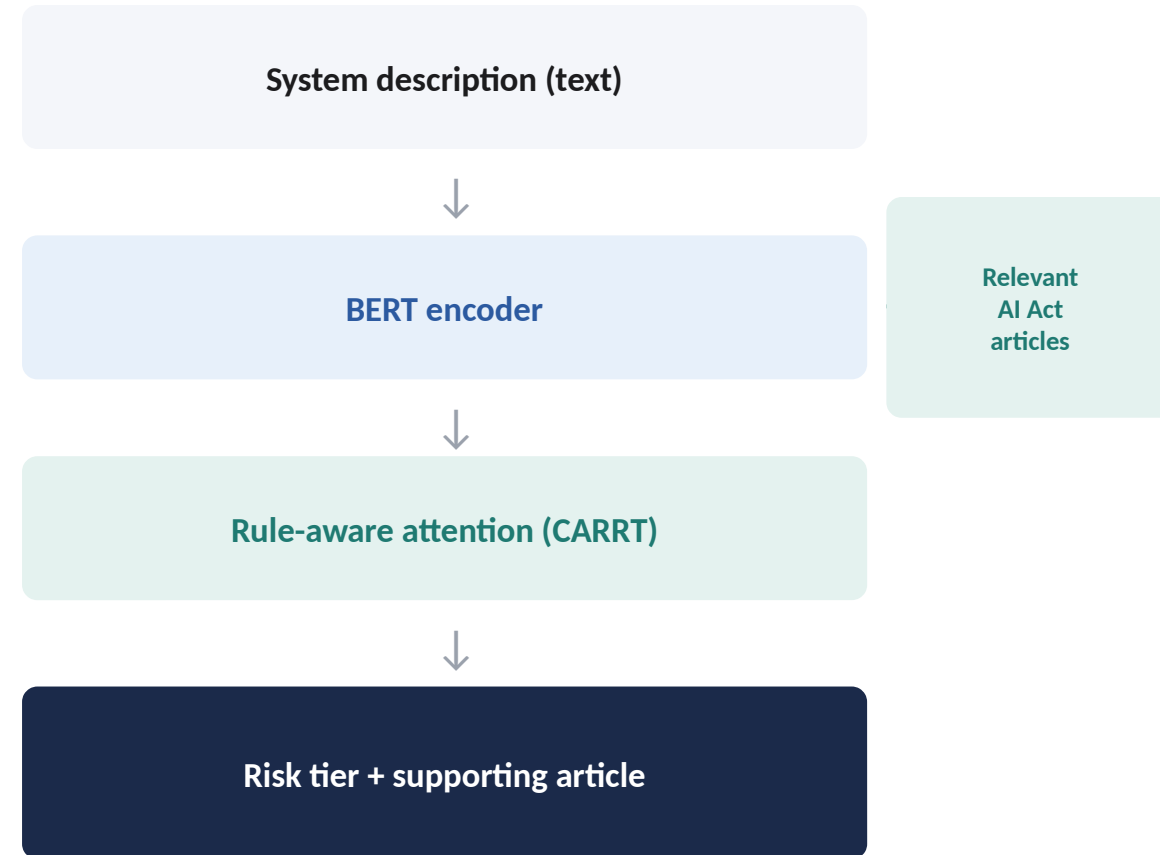
No prior work injects AI Act rules into transformer attention for risk classification

- **Focus area**

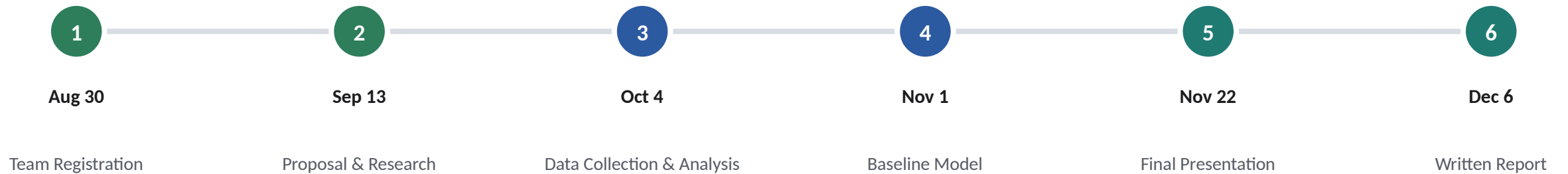
Prohibited & High-risk tiers — where missed classifications carry the most compliance risk

- **Builds on**

Rule + ML hybrid literature, applied for the first time with a modern transformer



Six stages, mapped to course deadlines



Where we are now

TF-IDF + XGBoost baseline is in progress. BERT fine-tuning is planned next. CARRT design begins once BERT results are in — giving us a clean comparison point before adding rule-injection.

What we expect to learn

Fine-tuning transformers for a specialized domain

How to adapt a general-purpose model like BERT to dense, high-stakes legal language — and what breaks when you do.

Designing a custom, rule-aware attention mechanism

How to inject external structured knowledge (AI Act articles) into a transformer's attention, rather than treating the model as a black box.

Building a multi-task classification + retrieval system

How to get one pipeline to both predict a risk tier and retrieve the specific article that justifies it.

Handling label ambiguity in imbalanced legal data

How to evaluate and improve a model when the hardest, rarest cases (Prohibited, High-risk) are also the most ambiguous to label in the first place.

Sources

1. Davvetas, A., Papademas, M., Ziouvelou, X., & Karkaletsis, V. (2026). AI Act Evaluation Benchmark: An Open, Transparent, and Reproducible Evaluation Dataset for NLP and RAG Systems. arXiv preprint arXiv:2603.09435.
2. European Union. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union.
3. Meghdadi, N., & Inkpen, D. (2024). uOttawa at LegalLens-2024: Transformer-based Classification Experiments. In Proceedings of the Natural Legal Language Processing Workshop 2024 (pp. 42-47). Association for Computational Linguistics.
4. Ali, F. A. R., Ahmed, S. A., Hasan, M. S., Sahi, S., Samir, M., Mahmod, S. A., & Ahmed, J. M. (2025). Automated Regulatory Compliance Verification for GDPR and ePrivacy Directives Using Hybrid Rule-Based Systems and Explainable AI Models. In Proc. 3rd International Conference on Cyber Resilience (ICCR). IEEE.
5. Tuteja, M., & Gonzalez Jucla, D. (2023). Long Text Classification using Transformers with Paragraph Selection Strategies. In Proceedings of the Natural Legal Language Processing Workshop 2023 (pp. 17-24). Association for Computational Linguistics.